Parallel Graph Algorithms to Remove Degenerate Features from Ice Sheet Meshes, and Determine Biconnectivity

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General Motivation

- Model Sea Level Rise
- Predict the future climate

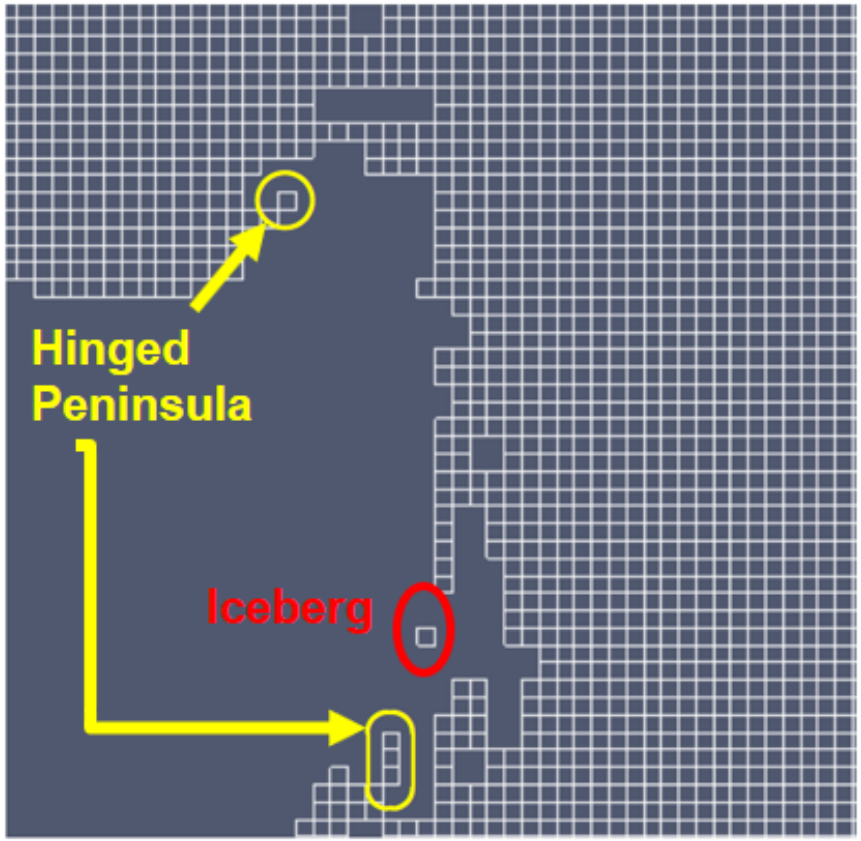


Specific Motivation

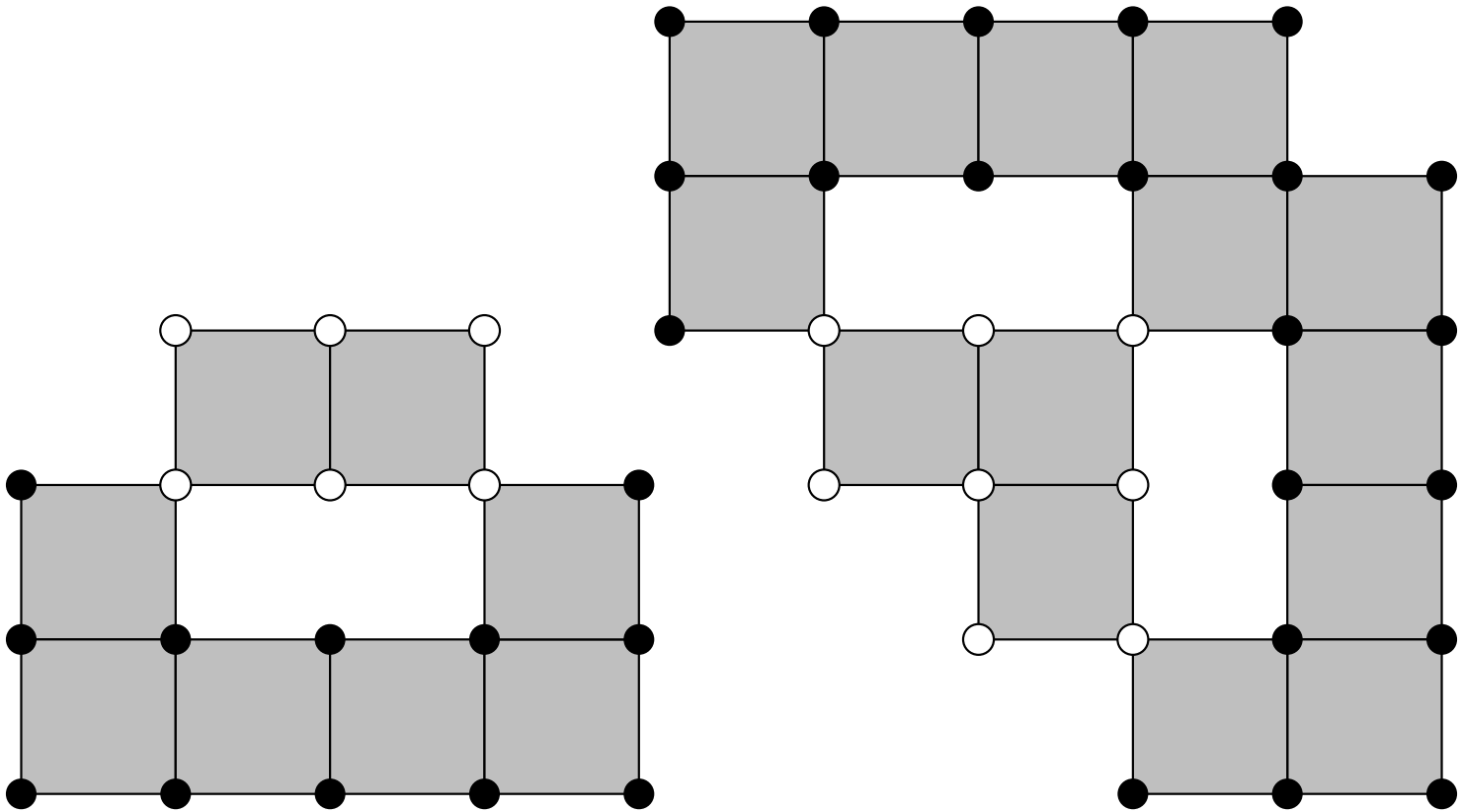
- Iterative solvers cannot handle certain mesh configurations
- These mesh configurations can arise during the simulation
- Current methods of removal[1] take too long to be used in-pipeline

Degenerate Features

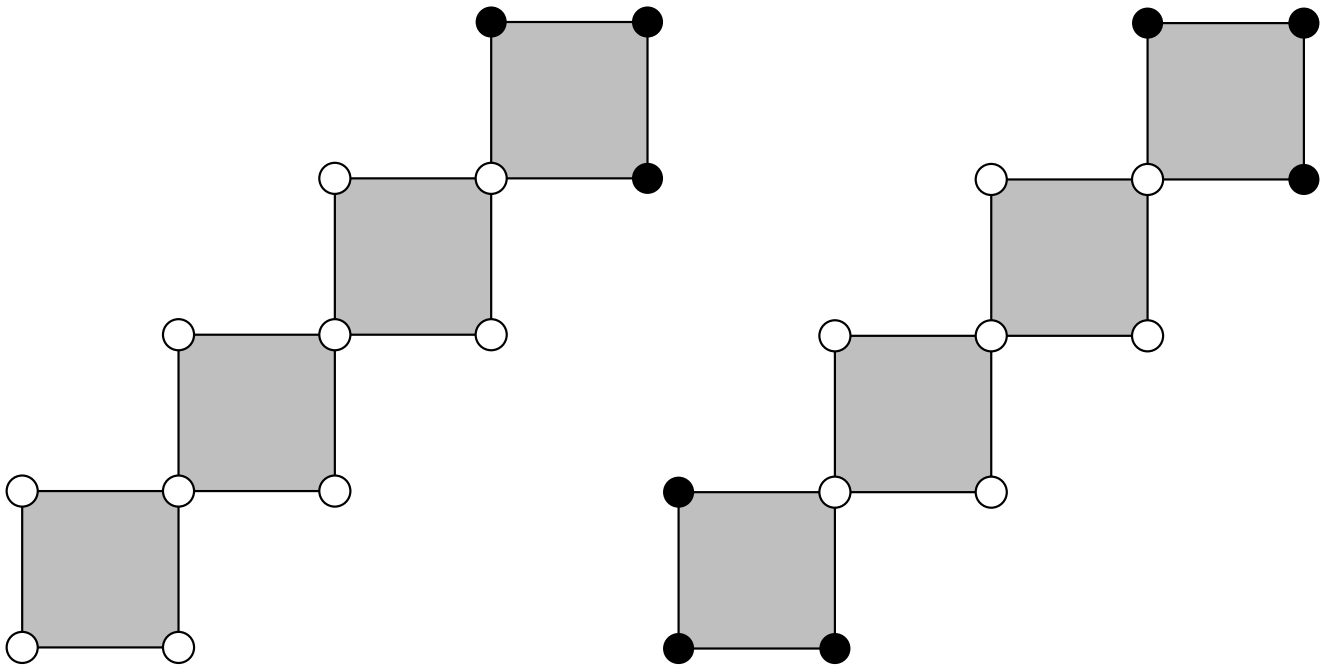
- Degenerate features can rotate or float freely
- We found unexpected features in the mesh



Unexpected Features



Unexpected Features



The Problem

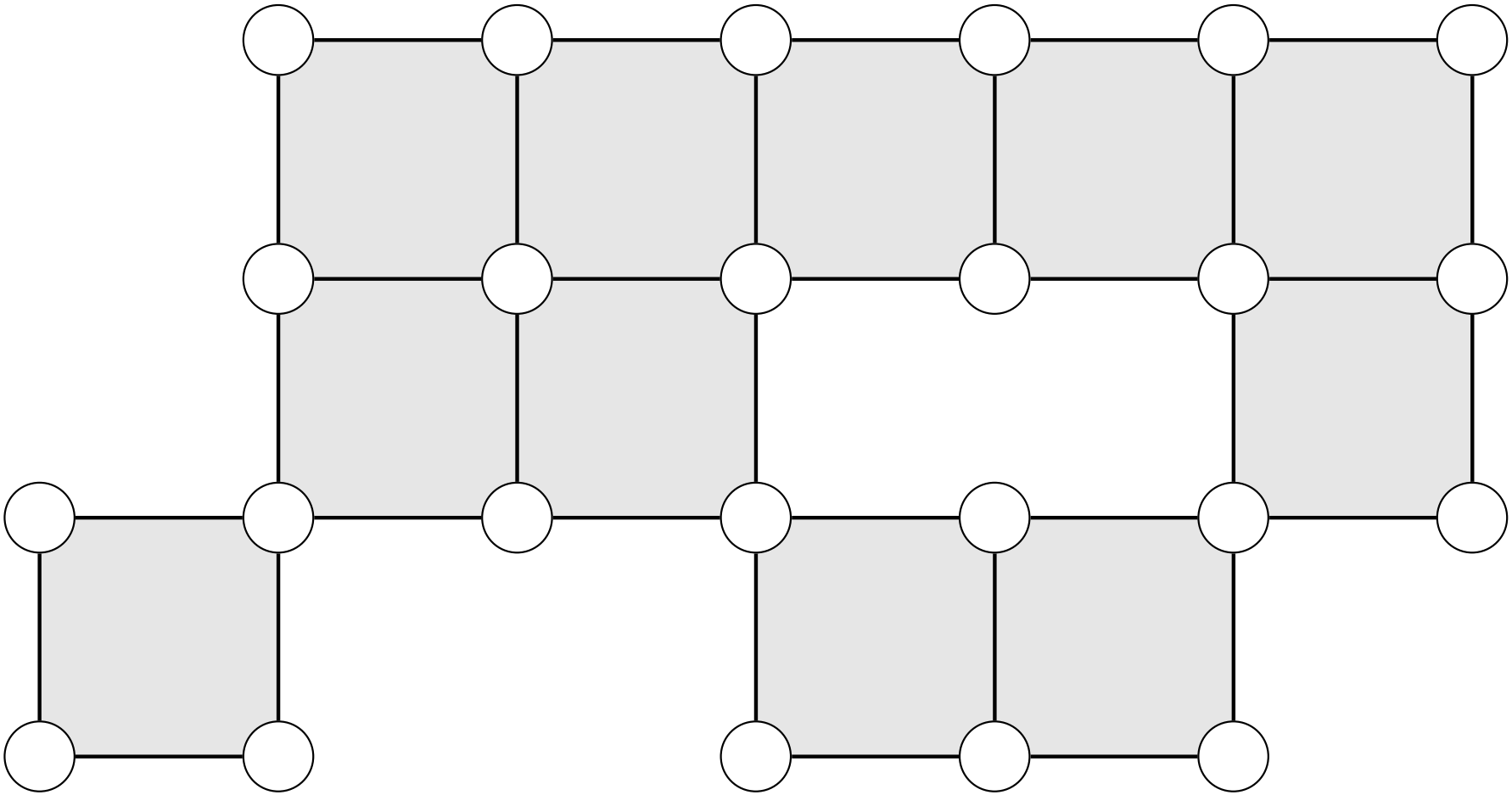
- Given:
  - Ice-sheet mesh
  - Grounding: Information about which vertices contact the ground
- We need to indicate degenerate features in the mesh

Our Approach

- Attempt to propagate grounding throughout the mesh
- Label propagation strategy with three components:
- Grounding Information
- Mesh-Based Articulation Point Heuristic
- Propagation Rules

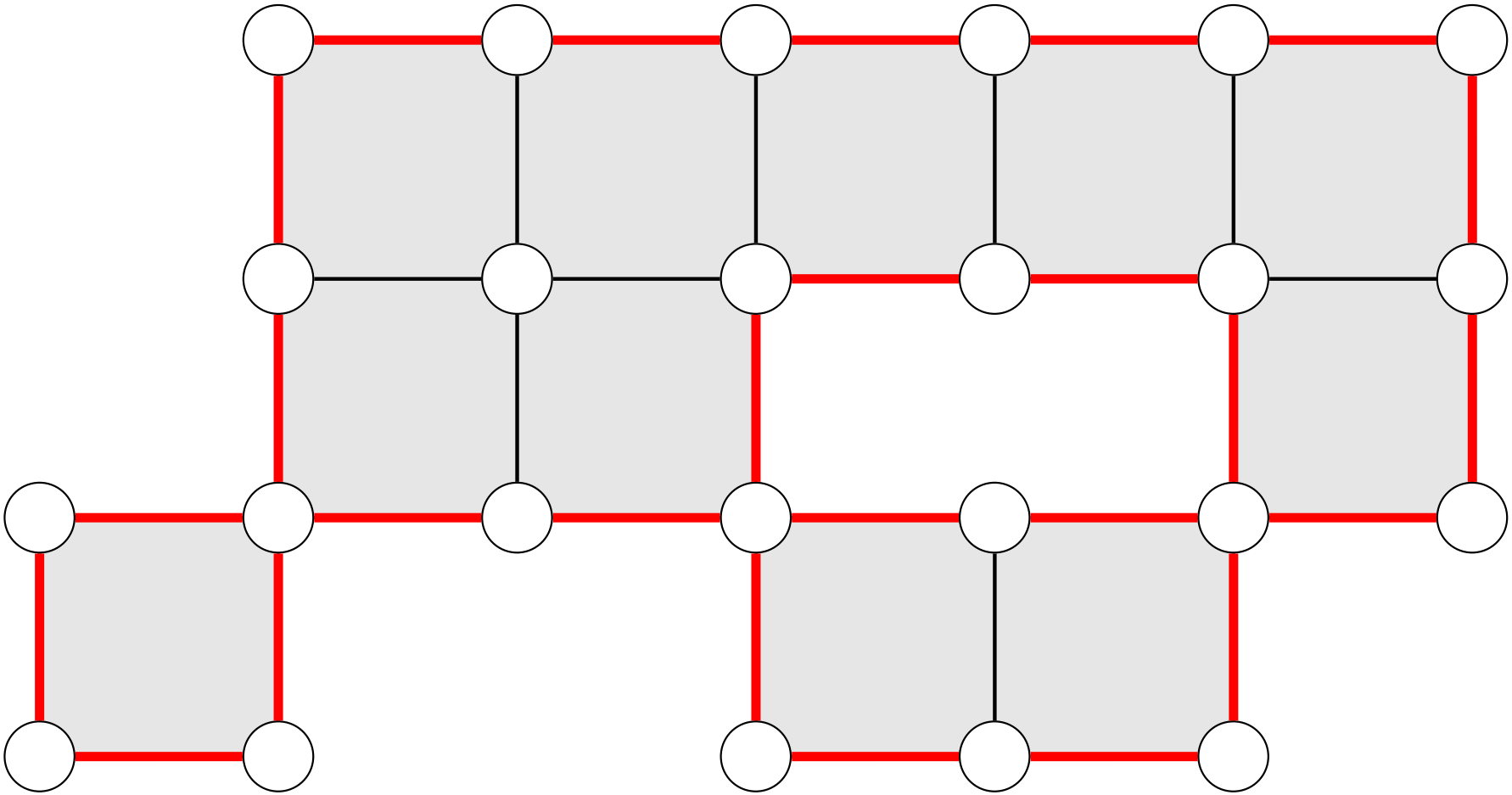
Mesh Heuristic

Initial Configuration, grounding not shown



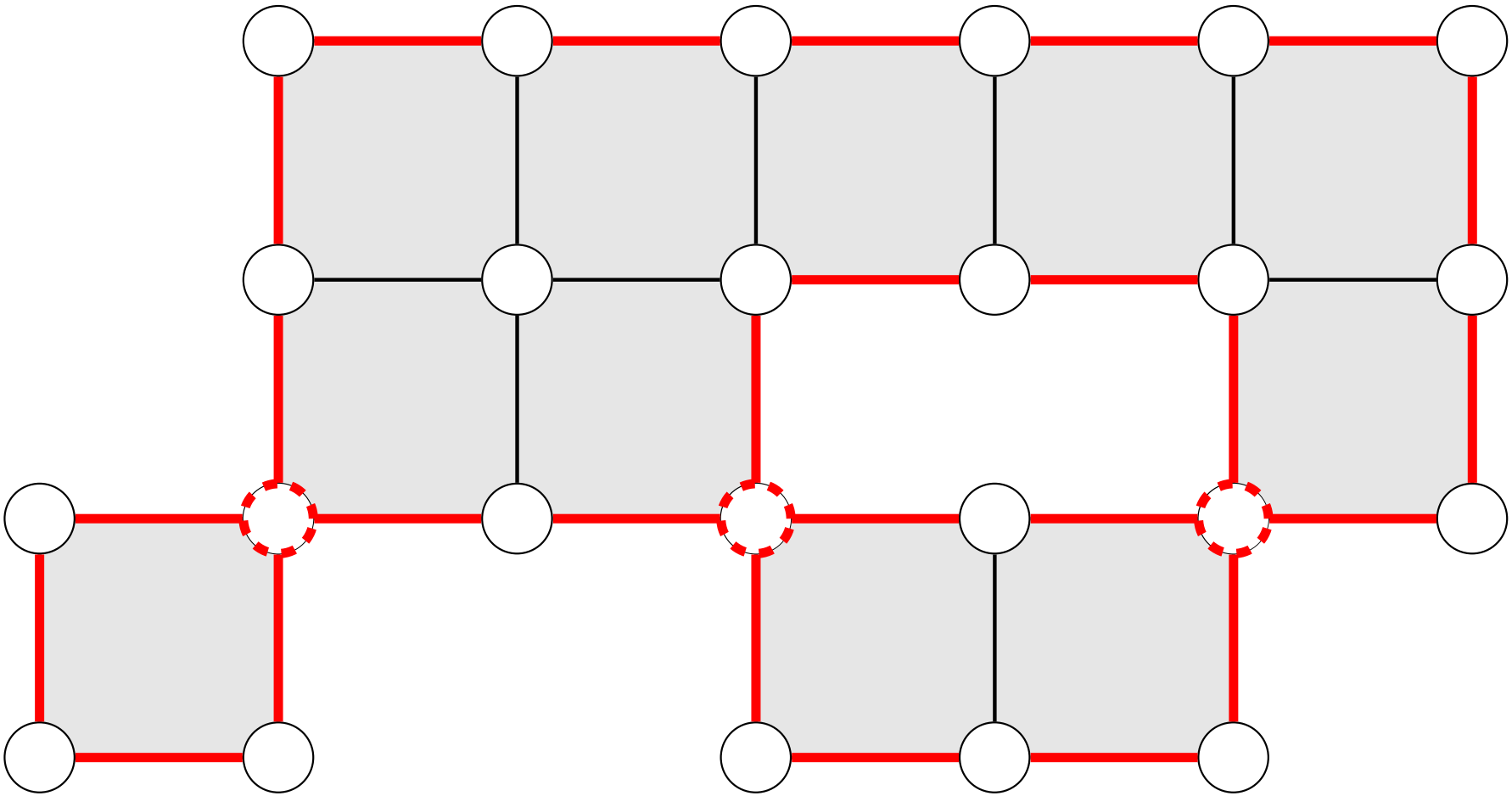
Mesh Heuristic

Note which nodes have more than 2 red edges



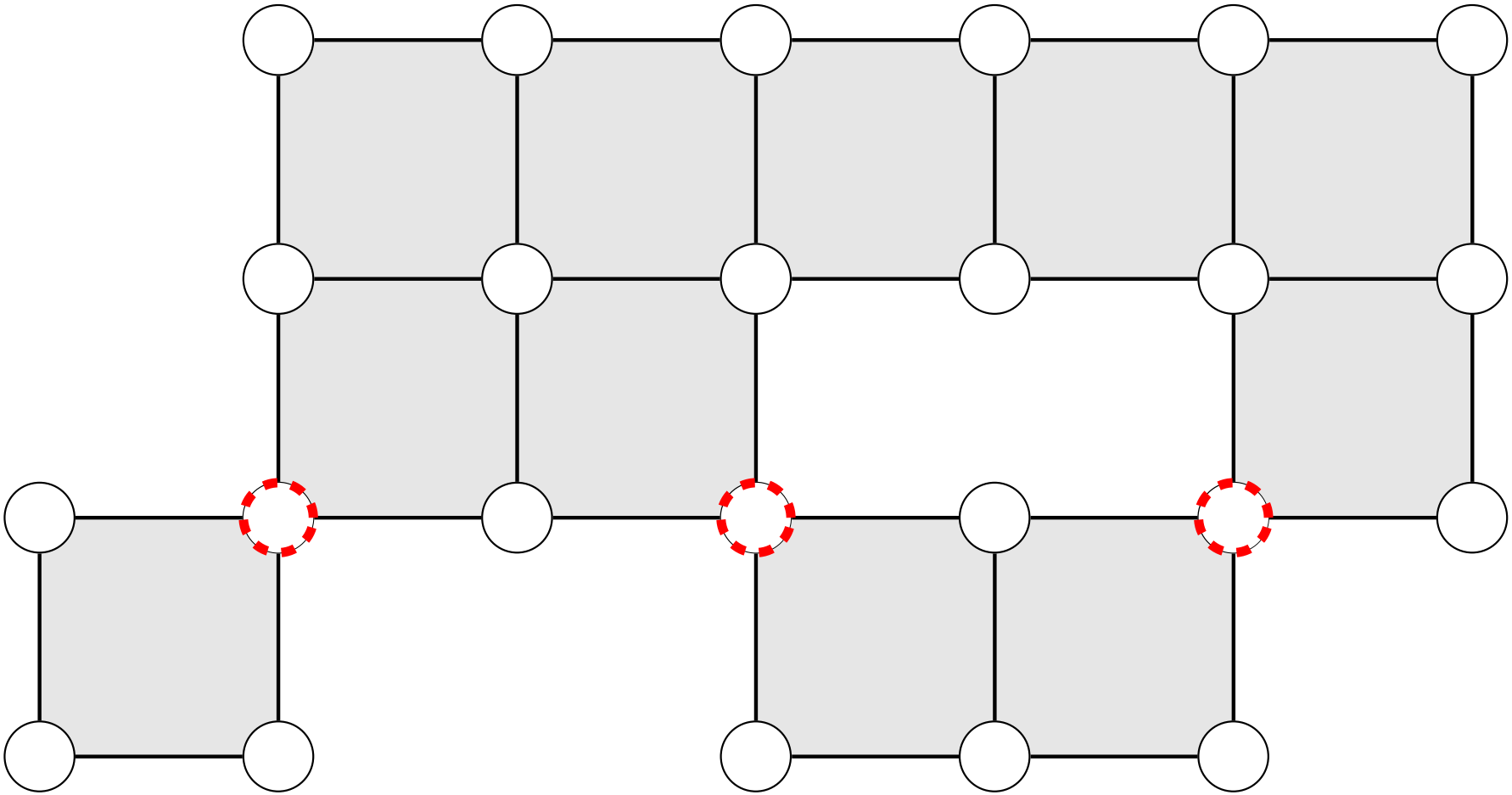
Mesh Heuristic

These nodes are Potential Articulation Points



Mesh Heuristic

These nodes are Potential Articulation Points



Labels

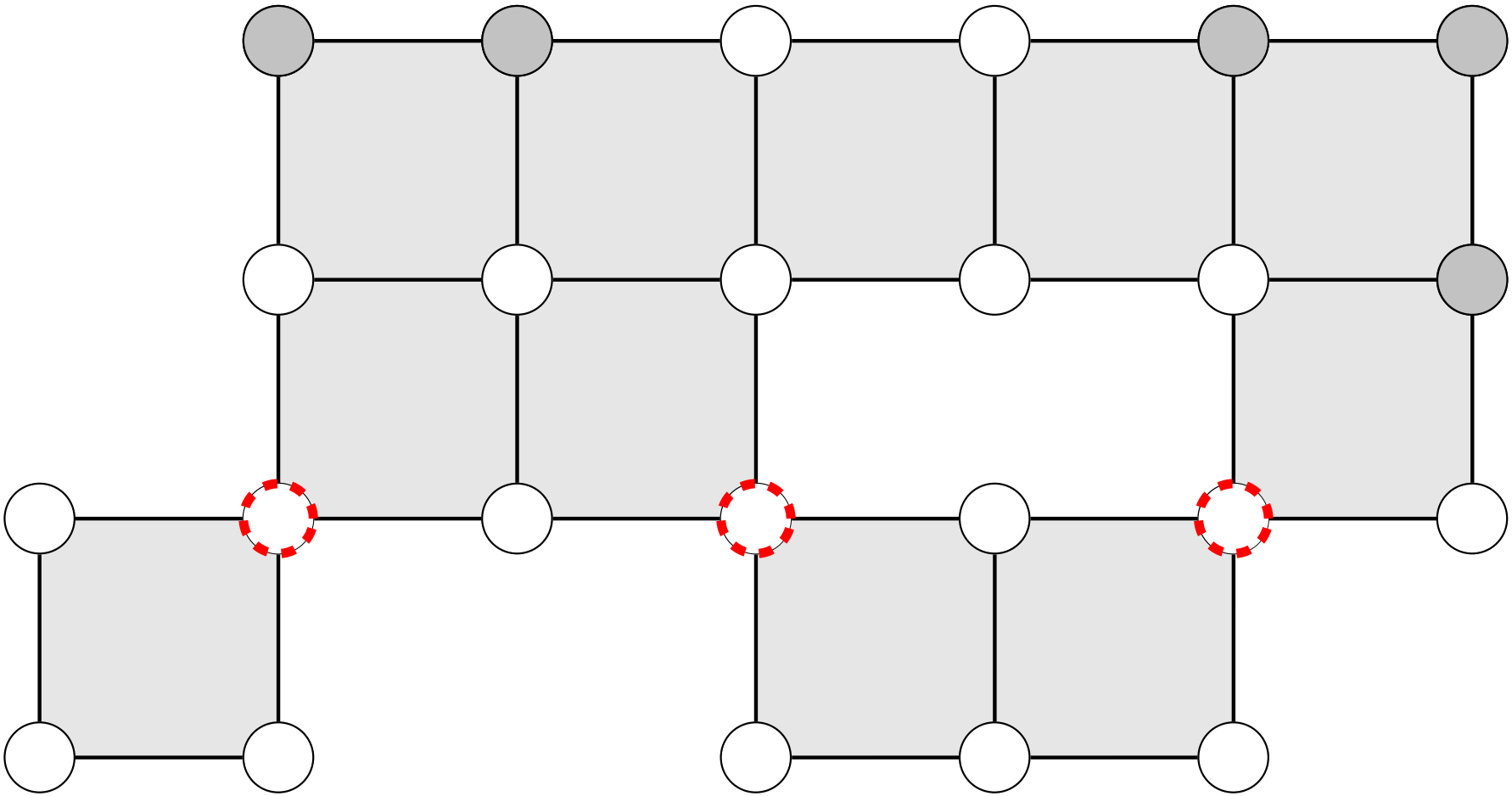
- Labels hold up to 2 global node (vertex) identifiers
- Nodes with labels having 1 identifier are colored grey
- Nodes with labels having 2 identifiers are colored black
- Nodes generally pass each other identifiers from their own labels

Propagation Rules

- Potential Articulation Points can only communicate to neighbors once
- If a Potential Articulation Point is full, it passes its own ID (not from its label)
- Any other vertex passes its entire label

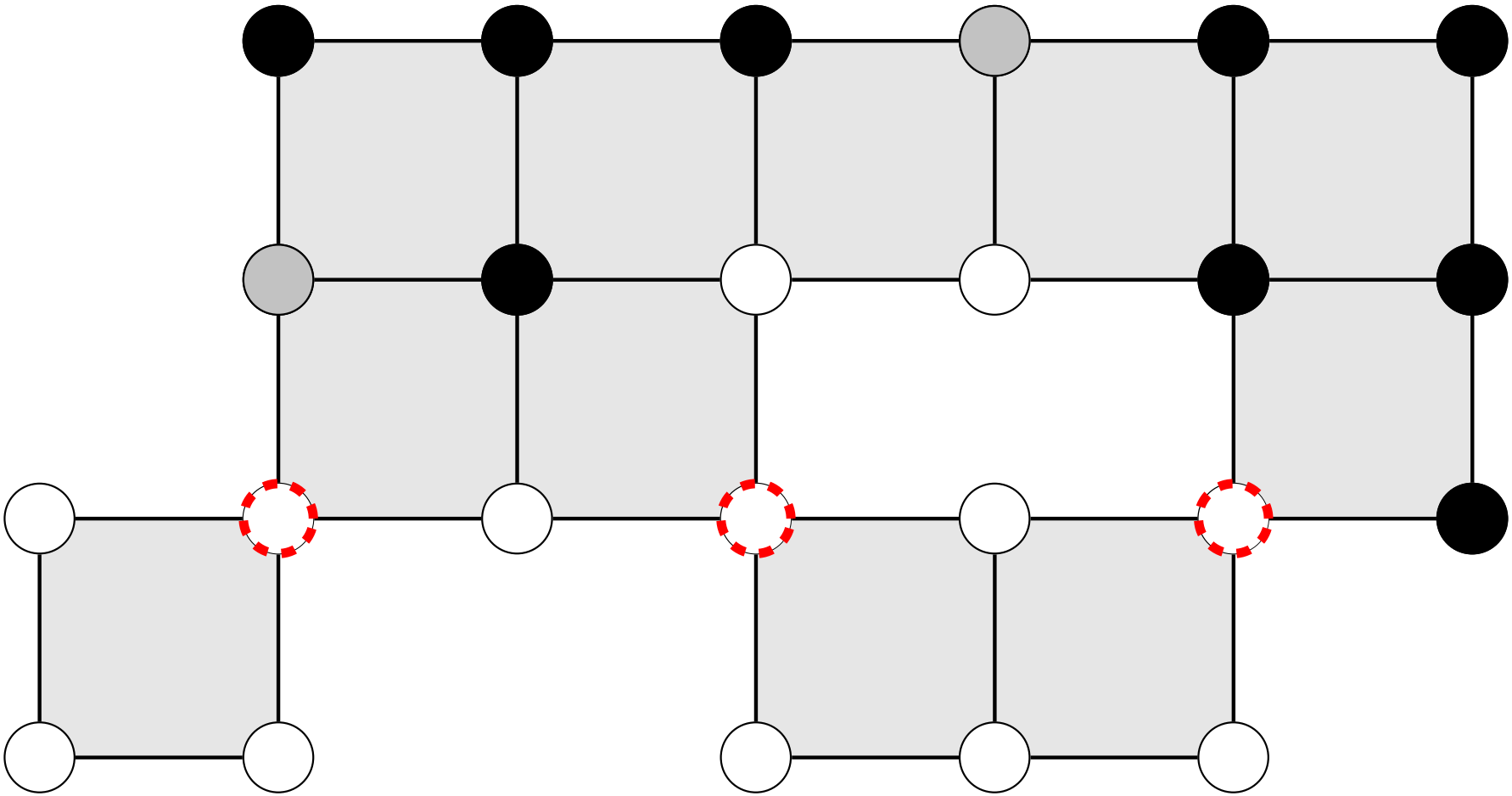
Propagation Example

Initial Configuration, gray nodes are touching the ground



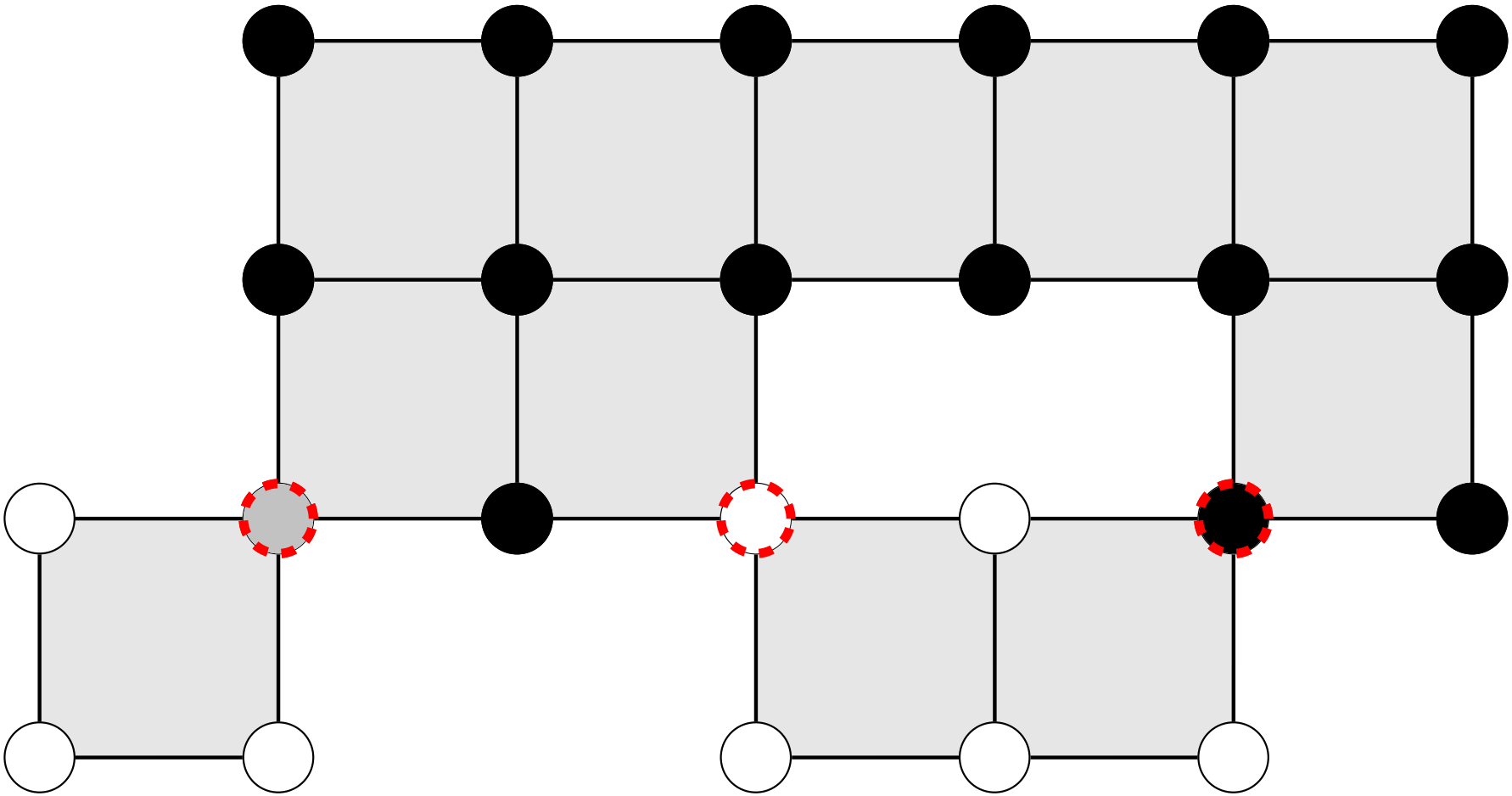
Propagation Example

After propagating from initially grounded nodes



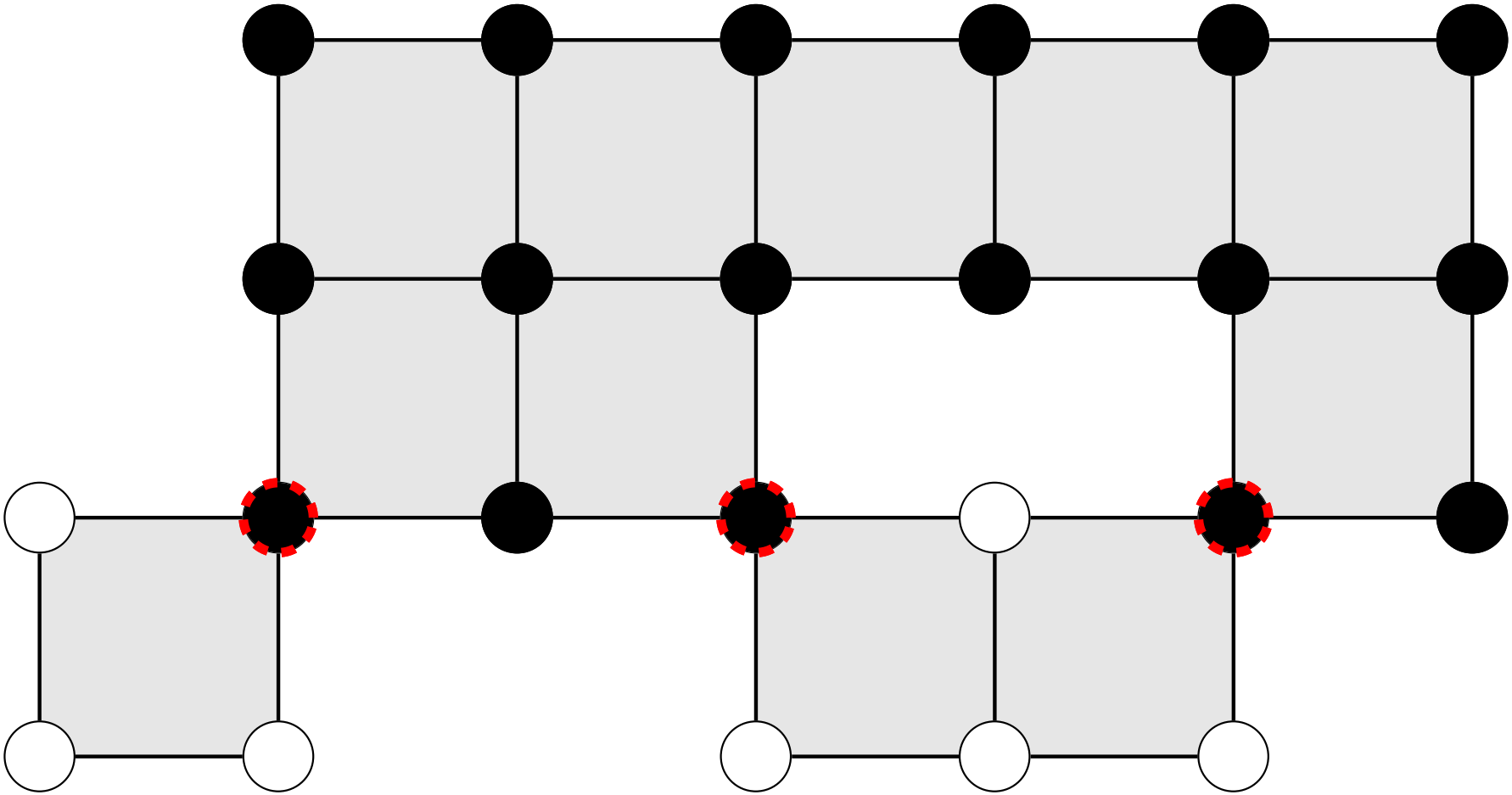
Propagation Example

After propagating from nodes that changed last time



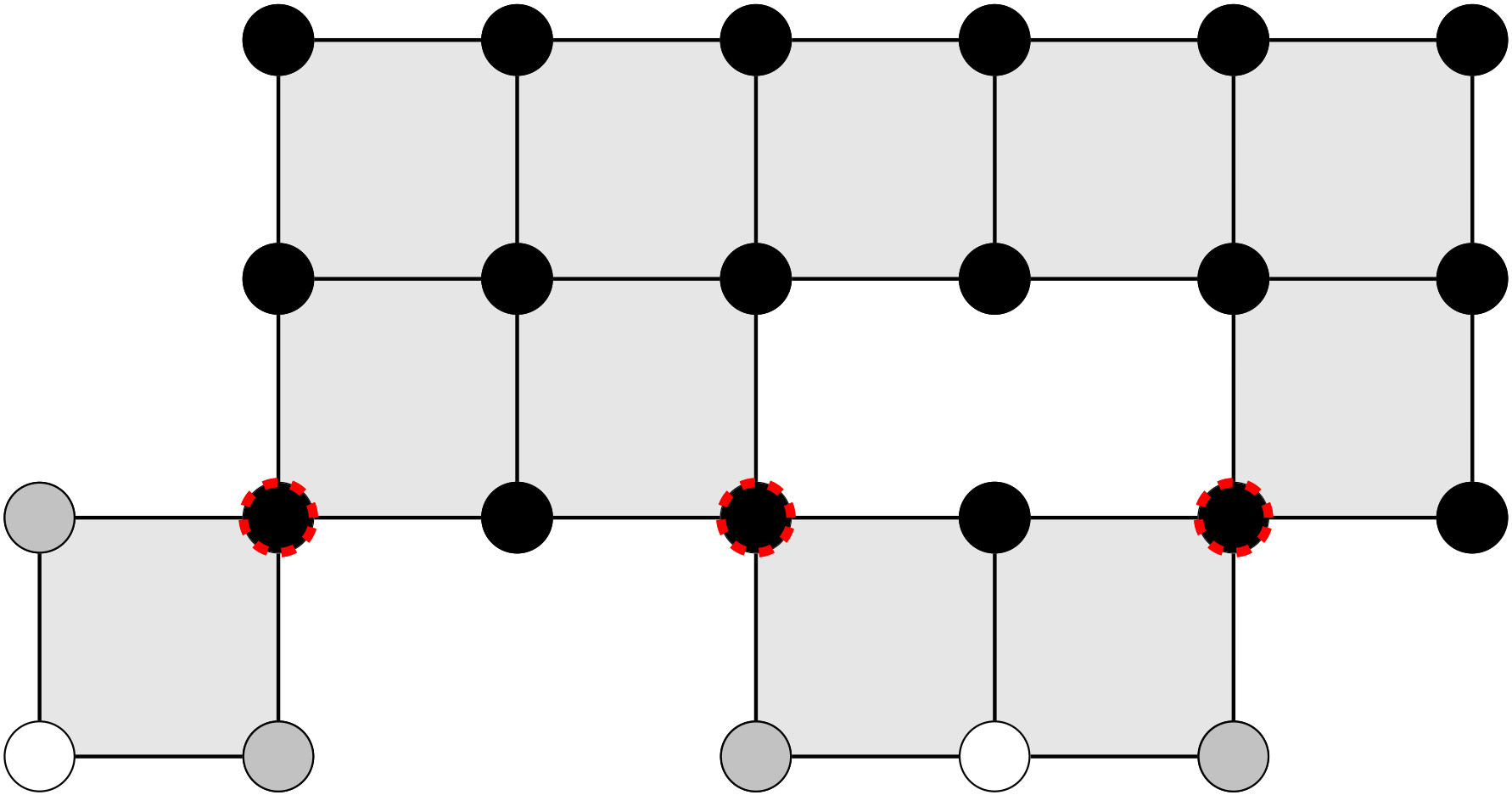
Propagation Example

We hold up propagation at the Potential Articulation Points



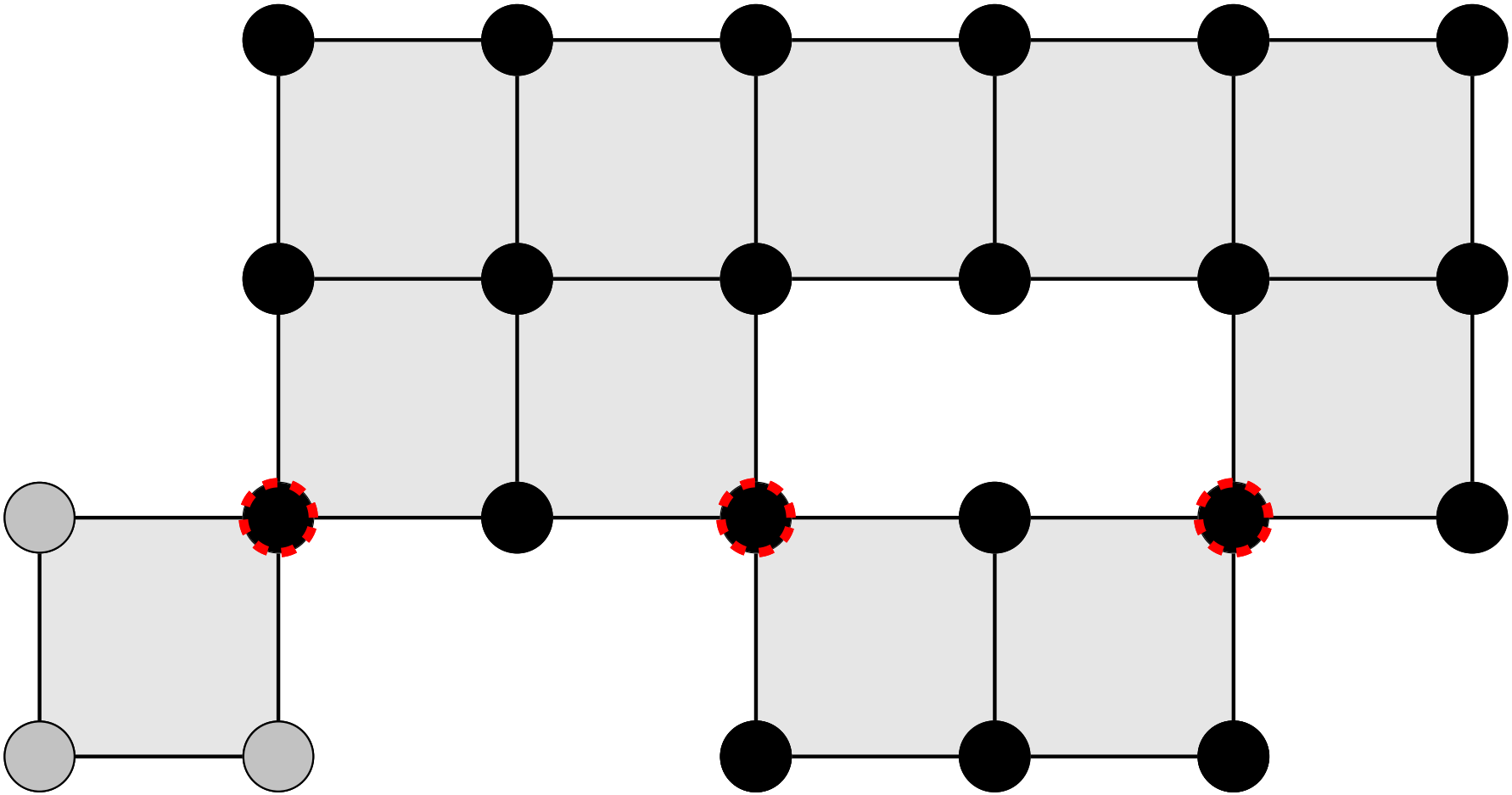
Propagation Example

Then propagate once they have a chance to gain more IDs



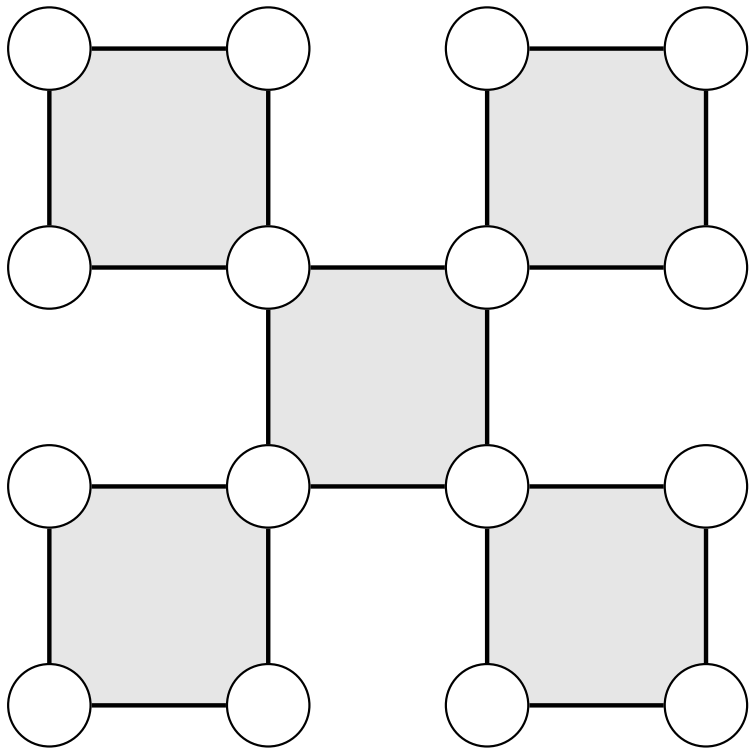
Propagation Example

Only degenerate features are grey or white

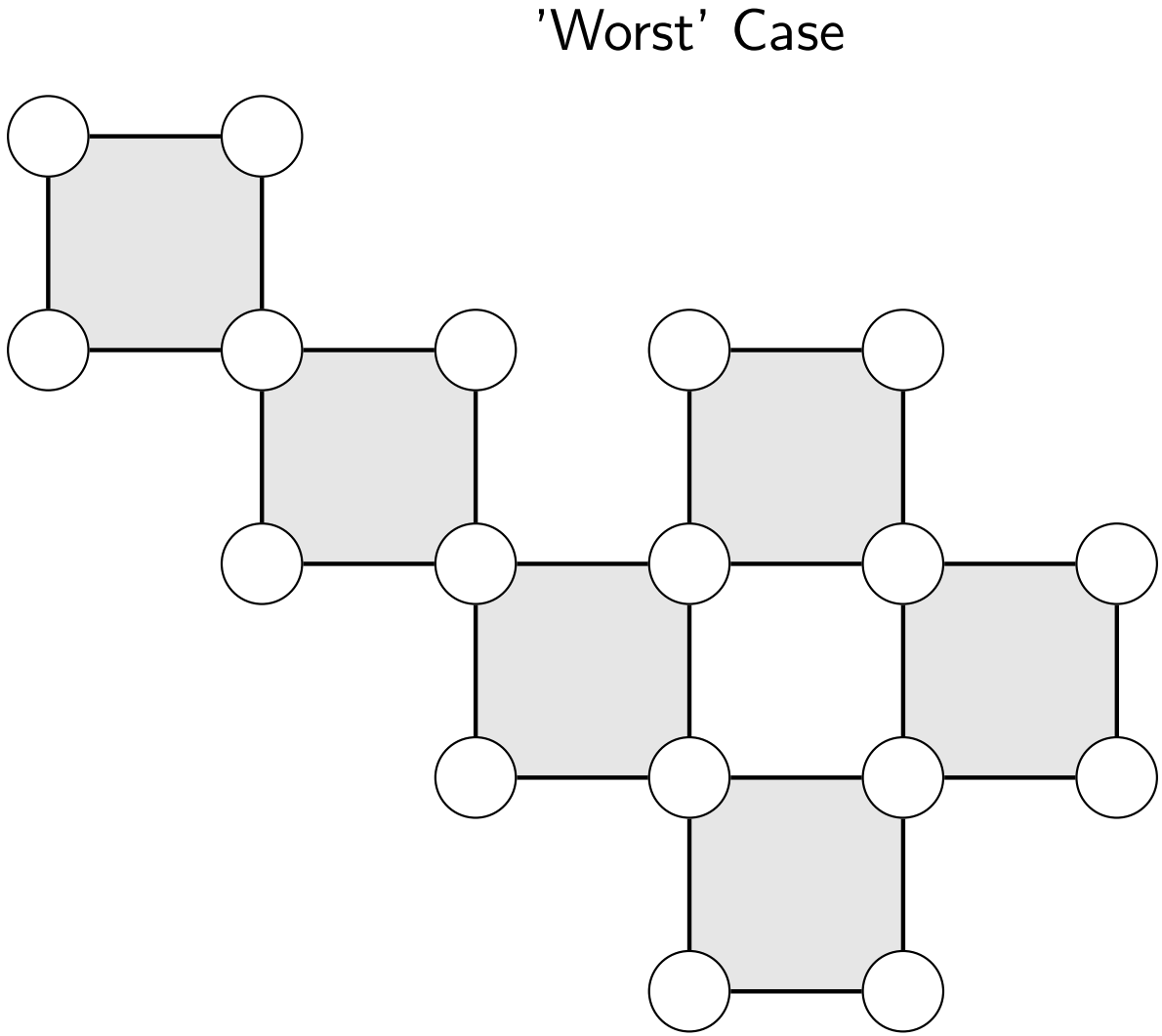


Corner Cases

'Tic Tac Toe' Case



Corner Cases



Biconnectivity - Background

- Biconnected components: subgraphs that remain connected if any single node is removed
- Articulation points can disconnect the original graph
- We can generalize our algorithm to find these components

Generalization

- Same propagation rules
- New articulation point heuristic
- New grounding: Any neighbors
- Iterative solution

Implementations

- Biconnectivity and Ice-sheet algorithms implemented with Trilinos
- MPI based implementation of the Ice-sheet algorithm
- Partial MPI implementation of Biconnectivity algorithm

Input Ice Sheets

- Input ice-sheets are relatively small

Resolution	Vertices	Elements	Edges	Potential
16km	52,465	51,087	204,348	21
8km	210,170	206,436	825,744	51
4km	841,346	831,173	3,324,692	174
2km	3,368,275	3,341,449	13,365,796	389
1km	13,479,076	13,413,766	53,655,064	606

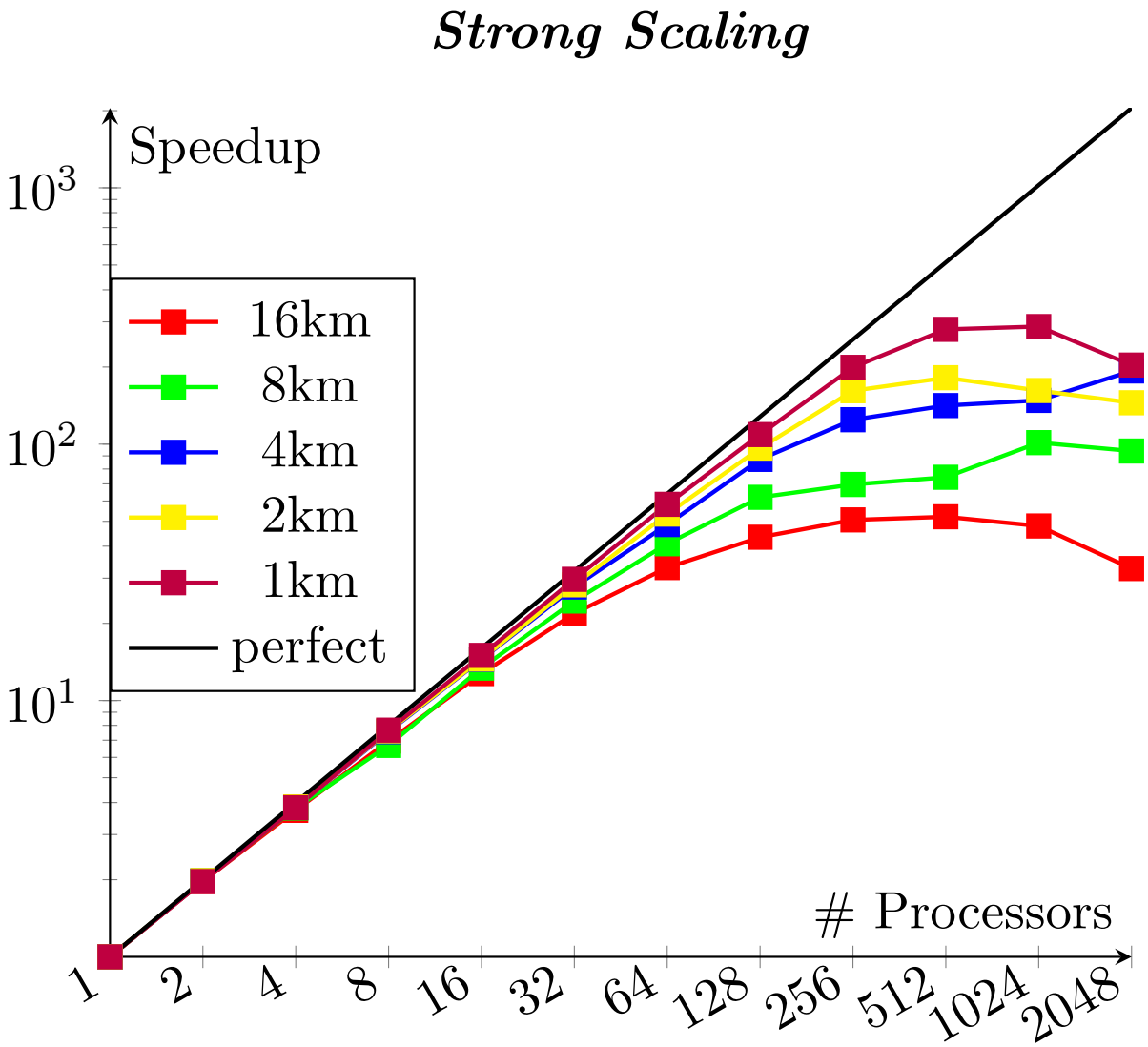
BCC Input Graphs

- Synthetic graphs
- Cycles with random edges, connected via articulation points
- Made sure no larger cycles were created when connecting cycles

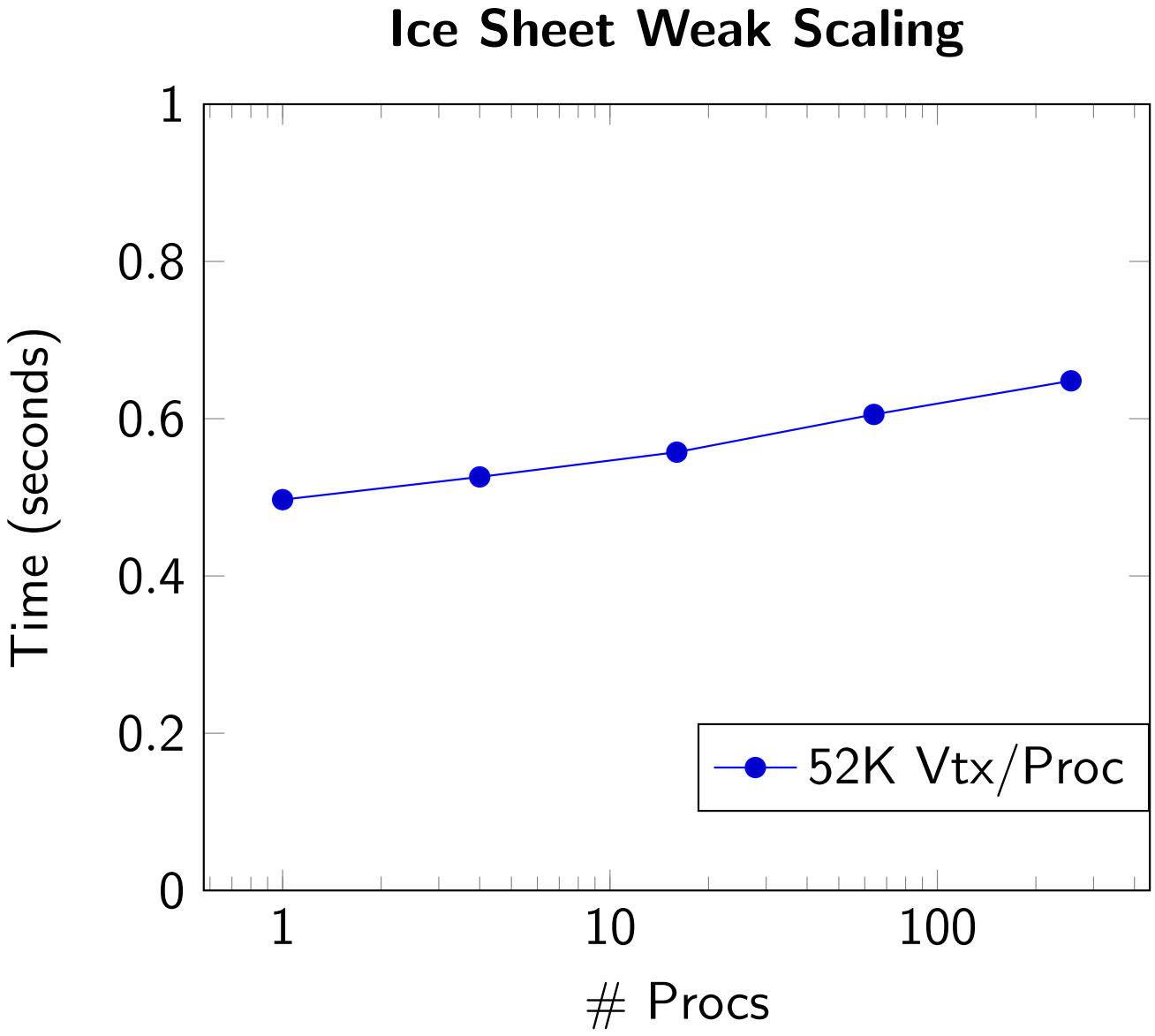
Testing Systems

- Amos - 5 rack Blue Gene/Q at RPI, 80 TB RAM total, 160 I/O nodes
- Blake - Advanced Architecture Platform at Sandia, 40 nodes, Dual-Socket Intel Xeon Platinum CPUs with dual AVX512 vector processing units

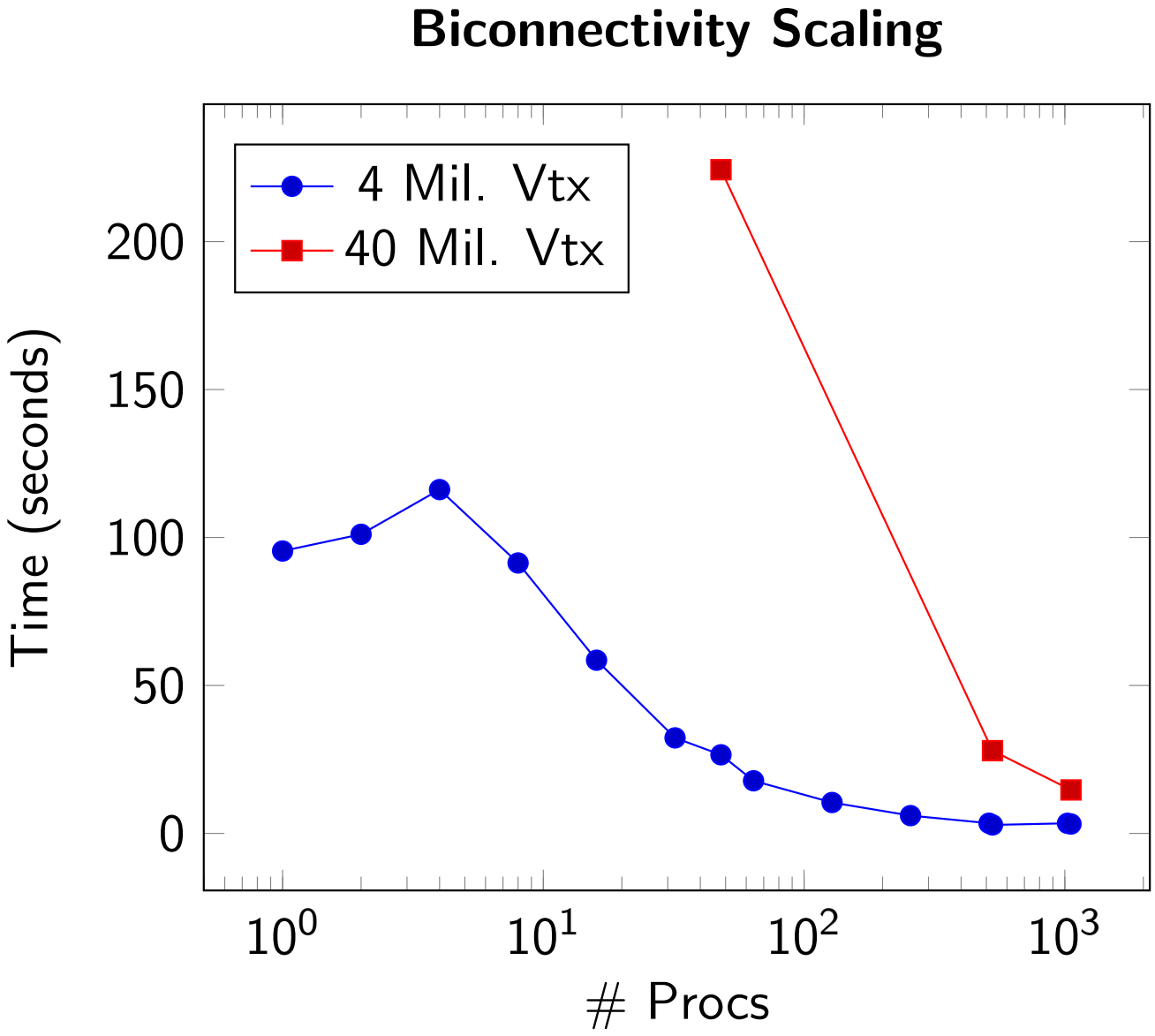
Results-AMOS



Results-AMOS



Results-Blake



Status and Future Work

- Finish MPI implementations
- Full Trilinos integration for simulation application

Conclusion

- Our label propagation algorithm solves the initial problem
- The algorithm is fast enough to be used in-pipeline
- We showed that it generalizes to solve biconnectivity

References



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